LCMatch: Layer Cross-Based Semi-Supervised Learning for Remote Sensing Scene Classification

Ruizhe Hu, Zuoyong Li, *Senior Member, IEEE*, Tao Wang, *Member, IEEE*, Sien Li, Yuanzheng Cai, Rong Hu, and George N. Papageorgiou, *Member, IEEE*

***Abstract*—We introduce LCMatch, a novel semi-supervised scene classification framework designed to enhance the performance of remote sensing image classification. Our method improves upon the existing FixMatch framework by incorporating a hierarchical structure for pseudo-label generation. The framework consists of three key modules: hierarchical cross-random combination (HCRC), adaptive weighting mechanism, and label alignment. These modules work synergistically to generate high-quality pseudo-labels, refining model predictions, and adaptively balancing the contributions of labeled and unlabeled data during training. In addition, we conduct extensive experiments on three widely used remote sensing datasets, including AID, UCMerced, and NWPU-RESISC45. Results demonstrate that LCMatch outperforms state-of-the-art semi-supervised learning (SSL) methods in terms of classification accuracy. Specifically, LCMatch exhibits robust performance even with a very limited number of labeled samples, also effectively handling class imbalance and distinguishing challenging categories.**

***Index Terms*—Consistency regularization, remote sensing, scene classification, semi-supervised learning (SSL).**

I. Introduction

REMOTE sensing imagery has become a crucial resource across various domains due to its ability to efficiently capture large-scale, high-resolution data. With rich spectral and spatial information, remote sensing images provide valuable insights for applications such as urban planning [1], precision agriculture [2], and natural disaster detection [3]. For instance, Li and Shao [4] proposed an object-based urban vegetation classification method using high-resolution aerial imagery, while investigating the spatiotemporal detection of urban villages [5]. In the field of agriculture, Huang et al. [6] discussed big data management in agricultural remote sensing, and Virnodkar et al. [7] applied remote sensing and machine learning techniques for crop water stress monitoring. Additionally, in disaster monitoring, Dong and Shan [8] reviewed the application of remote sensing in earthquake-induced damage detection, and Rudorff et al. [9] examined the impact of mining wastewater disasters on water turbidity. Consequently, research interest in remote sensing image processing has significantly increased in recent years, particularly in leveraging deep learning techniques to enhance automated analysis [10], [11], [12], [13].

Among various tasks, remote sensing scene classification is a fundamental problem in remote sensing image analysis, aiming to classify images into predefined semantic categories. Each image represents a specific object or scene, identified by semantic labels, which facilitate both human interpretation and automated machine analysis [14].

In addition to common issues such as high visual similarity and class imbalance, remote sensing scene classification presents several domain-specific challenges that make it distinct from generic image classification. First, remote sensing images often exhibit large intraclass variations due to differences in sensor modalities, spatial resolutions, viewing angles, and atmospheric conditions. Second, scene categories in remote sensing are usually defined by complex spatial layouts and contextual dependencies, rather than individual object appearance, which requires models to capture long-range structural and semantic information. Third, many remote sensing datasets suffer from limited annotation and significant label noise, particularly for fine-grained classes or underrepresented regions. Finally, variations across geographical regions and acquisition platforms introduce domain shifts that significantly impact model generalization and robustness. Therefore, methods that work well for generic image classification do not necessarily work well for remote sensing imagery.

Various deep learning-based approaches have been proposed to address these challenges. For instance, Wang et al. [15] introduced the Places205-VGGNet model for scene recognition, Chen et al. [16] developed a multibranch local attention network to enhance feature representation, and Zou et al. [17] proposed a feature selection approach based on deep learning. Furthermore, Cheriyadat [18] explored unsupervised feature

Received 21 June 2025; revised 21 August 2025 and 6 September 2025; accepted 7 September 2025. Date of publication 10 September 2025; date of current version 23 September 2025. This work was supported in part by NSFC under Grant 62471207, in part by Fujian NSF under Grant 2024J02029, and in part by the Science and Technology Project of Fuzhou Institute of Oceanography under Grant 2024F03. *(Corresponding authors: Tao Wang; Rong Hu.)*

Ruizhe Hu is with the School of Computer and Data Science, Minjiang University, Fuzhou 350108, China, and also with the College of Computer Science and Mathematics, Fujian University of Technology, Fuzhou 350118, China (e-mail: lkoor225@gmail.com).

Zuoyong Li, Tao Wang, and Yuanzheng Cai are with the School of Computer and Data Science, Minjiang University, Fuzhou 350108, China (e-mail: fzulzytdq@126.com; twang@mju.edu.cn; yuanzheng_cai@mju.edu.cn).

Sien Li is with the College of Computer and Data Science, Fuzhou University, Fuzhou 350108, China (e-mail: LlistenL@163.com).

Rong Hu is with College of Computer Science and Mathematics, Fujian University of Technology, Fuzhou 350118, China (e-mail: hurong0910@qq.com).

George N. Papageorgiou is with the SYSTEMA Research Center, European University Cyprus, 1516 Nicosia, Cyprus (e-mail: g.papageorgiou@euc.ac.cy).

Digital Object Identifier 10.1109/TGRS.2025.3608572

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:13:51 UTC from IEEE Xplore. Restrictions apply.

learning for aerial scene classification, while Yue et al. [19] introduced spectral–spatial latent reconstruction to improve the generalization of hyperspectral image classification. In the domain of semantic segmentation, Wu et al. [20] proposed the deep bilateral filtering network, whereas Bian et al. [21] investigated the fusion of local and global features to optimize high-resolution scene classification.

However, the rapid expansion of remote sensing datasets has posed significant challenges, particularly in data annotation. Traditional annotation methods heavily rely on human experts, making the process time-consuming, expensive, and prone to errors due to the high visual similarity and homogeneity of remote sensing images. These issues not only affect data quality but also increase model training complexity and reduce generalization ability. As a result, researchers have been actively exploring semi-supervised and unsupervised learning strategies to mitigate annotation challenges while maintaining classification accuracy.

Recent large-scale efforts such as SatlasPretrain [22] have highlighted the extreme diversity and complexity of remote sensing imagery. The dataset comprises 302 million labels across 137 categories and seven label types, covering a wide range of geospatial tasks from forest monitoring to urban infrastructure detection. It reveals that object sizes span several orders of magnitude, from kilometers to centimeters, and that many tasks require reasoning over time series or across heterogeneous sensors such as Sentinel-2 and NAIP. These characteristics impose significant annotation burdens and increase the cost of building large-scale supervised datasets, making semi-supervised learning (SSL) strategies particularly attractive for remote sensing applications.

Therefore, SSL [23], [24], [25], [26], [27], [28], [29] has emerged as an effective approach to address the scarcity of labeled data by leveraging abundant unlabeled data to improve model performance. In the domain of remote sensing scene classification, SSL has shown significant promise. For instance, Lee et al. [30] proposed a seminal method based on pseudo-labeling, where samples with confidence scores above a fixed threshold are selected as pseudo-labels for training, thus mitigating the impact of incorrect labels. However, the effectiveness of this method heavily depends on the predefined threshold, which may result in a decline in pseudo-label quality during the later stages of training.

Similarly, the FixMatch [31] framework integrates consistency regularization with pseudo-label generation. In this approach, pseudo-labels are generated using weak augmentation and used to train strongly augmented samples. FixMatch has demonstrated considerable success across various domains.

In SSL, a single model often depends on erroneous pseudo-labels for training, leading to the gradual accumulation of errors and overfitting as training progresses. To address this issue, we propose the Layer Cross method, which constructs a composite model for pseudo-label prediction by randomly combining the original model with the exponential moving average (EMA) model [32]. This strategy effectively mitigates the risk of overfitting to incorrect pseudo-labels during training, thereby enhancing the model's stability and robustness.

As training progresses, the predictions of the model tend to concentrate on a few majority categories with higher accuracy, causing predictions for other categories to decrease or become erroneous. This issue is particularly prevalent in fine-grained classification tasks, where certain minor categories may receive insufficient training, negatively affecting overall performance. To resolve this problem, we propose the Label Alignment method that smooths the distribution of pseudo-labels to ensure the model can handle all categories equitably during training. This is especially beneficial for categories with poorer performance, allowing the model to maintain some learning ability for them. This method alleviates class imbalance and improves the model's ability to predict minority classes, enhancing overall classification performance.

Also, as training progresses, the influence of labeled data diminishes because the labeled samples are limited, and the model becomes less reliant on them. At this stage, training increasingly depends on pseudo-label generation, with unlabeled data playing a more prominent role. To further enhance the effectiveness of pseudo-labels, we introduce an adaptive weighting module. This module dynamically adjusts the weight of unlabeled data based on the pseudo-label utilization rate and training phase. It maximizes the impact of high-confidence pseudo-labels while reducing the negative influence of low-confidence pseudo-labels. This dynamic weighting mechanism significantly improves the stability of training and the final accuracy of the model.

It is important to note that remote sensing scene classification presents some unique challenges. On the one hand, remote sensing images exhibit high visual similarity, particularly in fine-grained classes (e.g., distinguishing between forests and orchards), making it difficult for models to learn discriminative features during early training stages. On the other hand, remote sensing datasets often suffer from class imbalance, further exacerbating the accumulation of erroneous pseudo-labels and hindering overall performance.

To address these challenges, we propose a novel SSL framework, LCMatch, which optimizes the traditional FixMatch method to better adapt to remote sensing scene classification. The key contributions of LCMatch are summarized as follows:

1) *Layer-Wise Cross Random Combination Prediction:* LCMatch integrates different layers from both the original and EMA models to construct a composite model for predicting pseudo-labels for weakly augmented samples. This random combination strategy mitigates overfitting to incorrect pseudo-labels, leading to more stable and reliable predictions, especially in scenarios with high visual similarity among remote sensing images.
2) *Adaptive Weighting Module:* An adaptive weighting module is introduced to dynamically adjust the loss weights of unlabeled data during training. Unlike fixed-weight schemes, this method adjusts weights based on the model's learning state, ensuring the full utilization of high-confidence pseudo-labels while minimizing the impact of low-confidence ones.
3) *Label Alignment Method:* A redesigned label alignment technique is employed to address class imbalance. By smoothing and calibrating the pseudo-label distribution,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:13:51 UTC from IEEE Xplore. Restrictions apply.

LCMatch ensures more balanced predictions across all classes, improving performance, particularly for minority classes.

4) *Effective Performance on Remote Sensing Datasets:* LCMatch has shown remarkable results in overcoming the challenges inherent in remote sensing data, such as high visual similarity and class imbalance. The model's robustness and effectiveness are proved through experiments conducted on three public remote sensing datasets: WHU-RS19, AID, and UCM.

By leveraging the aforementioned innovations, LCMatch effectively utilizes unlabeled data while overcoming the limitations of existing methods, such as overfitting and the accumulation of errors. Experimental evaluations on multiple remote sensing datasets substantiate the robustness and effectiveness of the proposed approach, yielding substantial performance gains, particularly in class-imbalanced and fine-grained scene classification tasks.

Despite the promising performance of existing semi-supervised methods such as FixMatch and Mean Teacher, their core limitations still hinder their application to remote sensing scenarios. FixMatch heavily relies on high-confidence thresholding of single-view predictions, which can propagate early-stage errors due to confirmation bias. Mean Teacher, while introducing temporal ensembling, does not account for internal feature diversity and lacks structural perturbation to improve pseudo-label reliability. These issues are particularly detrimental in remote sensing datasets where interclass differences are subtle and class distributions are often skewed.

LCMatch addresses these gaps from both architectural and optimization perspectives. The proposed hierarchical cross-random combination (HCRC) introduces a novel feature-level stochastic fusion strategy, injecting structural diversity and improving the semantic expressiveness of pseudo-labels. This approach not only enhances robustness against noisy labels but also serves as a form of implicit regularization during training. Furthermore, the dynamic adaptation of loss weights and class-wise alignment enables the model to better capture minority classes while ensuring stability as training progresses. These design choices collectively endow LCMatch with stronger generalization and error resilience, particularly suited for challenging remote sensing classification tasks.

The remainder of this paper is organized as follows. Section II reviews related work. Section III presents the proposed framework. Section IV provides experimental setup and results, including comparative evaluations. Section V discusses the findings and provides the main concluding remarks.

II. Related Work

This section reviews previous studies related to our work, focusing on remote sensing image scene classification and SSL.

A. Remote Sensing Image Scene Classification

Remote sensing image scene classification methods can be broadly categorized into two main approaches: CNN-based methods [33] and GAN-based methods [34], [35], [36].

Among CNN-based approaches, RSFJR [37] integrates spatial- and frequency-domain features within the feature extraction pipeline, enhancing feature robustness and discrimination. ADFF [38] dynamically fuses features from different layers, adaptively combining multiscale semantic information to improve classification accuracy. TRS [39] effectively combines the local feature extraction capabilities of CNNs with the global feature modeling strengths of Transformers, utilizing multihead self-attention mechanisms to significantly enhance the classification of complex remote sensing scenes. Classical CNN models, such as VGGNet [40] and ResNet [41], which have demonstrated exceptional performance in vision tasks, also serve as a solid foundation for remote sensing scene classification.

For GAN-based methods, MARTA GANs [42] employ multilayer feature matching to align features of generated and real samples across different semantic levels, improving the learning of domain-invariant representations. Attention GAN [43] incorporates an attention mechanism in the discriminator to focus on discriminative regions in remote sensing images, thereby enhancing feature representation and classification accuracy. Sifting-GAN [44] introduces a filtering mechanism to generate diverse high-quality annotated samples, ensuring that only high-quality generated samples are used for training, which significantly improves the generalization capability of the classification models.

Recently, the increasing demand for large-scale and fine-grained scene understanding has driven the development of benchmark datasets and more robust classification frameworks. For example, the MEET dataset [45] provides a million-scale benchmark for fine-grained geospatial scene classification using zoom-free remote sensing imagery. It highlights the necessity for models capable of generalizing across diverse categories and spatial contexts. Additionally, the hierarchical one-class detection method [46] proposed in addresses the challenge of separating foreground and background classes in hyperspectral image classification, offering valuable insights for handling imbalanced and noisy label distributions—issues that are also highly relevant in our scenario.

B. Semi-Supervised Learning

SSL aims to improve model performance by effectively leveraging unlabeled data. The core paradigms in SSL are consistency regularization, pseudo-labeling, and graph-based methods, which represent three fundamental approaches, each contributing to the efficient utilization of unlabeled data from distinct perspectives.

Consistency regularization is grounded in the manifold or smoothness assumption, asserting that the model's predictions should remain invariant under small perturbations of the input data. Representative methods include: Π Model [47], which applies random augmentations to the same sample and minimizes the discrepancy between the two predictions; Mean Teacher [48], which utilizes a student–teacher framework where the teacher model is updated via EMA of the student model's parameters to enforce consistency; Virtual Adversarial Training [49], which introduces adversarial perturbations to identify the most disruptive directions and counteracts

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:13:51 UTC from IEEE Xplore. Restrictions apply.

them through consistency constraints; and Unsupervised Data Augmentation [50], which integrates high-quality augmentation techniques, such as RandAugment, to further enhance consistency training.

Pseudo-labeling methods generate pseudo-labels for unlabeled data by assuming that the model's high-confidence predictions are generally correct. Key approaches include: pseudo-labeling [30], which selects the class with the highest confidence as the pseudo-label; SimCLR v2 [35], which extends contrastive learning by incorporating pseudo-labels during the fine-tuning stage, allowing the model to leverage unlabeled data for better feature representation learning and downstream task performance; and Noisy Student [51], which iteratively trains a student model using pseudo-labeled data produced by a teacher model, while adding noise to improve robustness and generalization. These approaches utilize high-confidence predictions to expand the labeled dataset, effectively integrating unlabeled data into the training process.

Graph-based methods exploit graph structures to model the geometric relationships among data points, propagating label information from labeled to unlabeled data under the manifold assumption. Notable techniques include: label propagation, which updates each node's label based on its neighbors in the graph; graph convolutional networks (GCNs) [36], which perform convolution operations on graph structures to aggregate neighborhood information and refine feature representations; and Laplacian regularization, which constrains adjacent nodes in the graph to maintain similarity in the feature space.

Hybrid approaches that integrate multiple paradigms often achieve superior results compared to individual techniques. For example, FixMatch [31] enforces consistency between weakly augmented pseudo-labels and strongly augmented data to optimize the model, while FlexMatch [52] dynamically adjusts confidence thresholds to accommodate the varying learning pace across classes, thereby improving pseudo-label quality. These methods exemplify the synergy of combining consistency regularization and pseudo-labeling to address challenges in SSL, enhancing both robustness and effectiveness.

However, while these methods have shown strong results in general SSL benchmarks, they face limitations when applied to remote sensing data. For instance, the strong domain shifts, complex spatial structures, and class imbalance in remote sensing datasets often result in low-quality or biased pseudo-labels. Moreover, fixed confidence thresholds used in these approaches may not adapt well to the dynamic and imbalanced pseudo-label distributions encountered in practice.

To tackle these issues, recent efforts in hyperspectral and remote sensing domains have introduced attention-based and modality-enhanced techniques [53]. For example, attention-based feature processing methods [54] have been proposed to improve cross-domain hyperspectral classification by dynamically enhancing domain-invariant features, while spatial–spectral fusion networks jointly exploit spectral and spatial cues to improve classification under few labeled samples. These approaches demonstrate the importance of domain-specific architectural adaptation in SSL.

While existing SSL methods, such as FixMatch and Mean Teacher, have shown success in generic benchmarks, their reliance on single-view predictions or fixed temporal ensembling limits their adaptability to the structural complexity of remote sensing images. LCMatch introduces a novel hierarchical cross-layer pseudo-labeling strategy, which integrates multilevel representations from the student and EMA teacher models. This mitigates confirmation bias and enhances semantic consistency in pseudo-labels. In addition, LCMatch incorporates a label alignment module to dynamically calibrate pseudo-label distributions under class-imbalanced settings, and an adaptive weighting mechanism to control gradient propagation based on pseudo-label confidence. These architectural and optimization-level enhancements distinguish LCMatch from prior SSL approaches and provide better alignment with the unique characteristics of remote sensing scene classification.

III. Method

In this section, we first provide a brief introduction to the FixMatch framework used in Section III-A, followed by a detailed explanation of the architectural improvements in Section III-B. Then, in Sections III-C and III-D, we introduce two innovative modules of the proposed LCMatch framework in detail, namely the adaptive weight module and the label alignment module. A high-level overview of our method is presented in Fig. 1.

A. Basic Framework

In existing SSL frameworks, our research is primarily inspired by the FixMatch method. Formally, a remote sensing image is represented as $x_i \in \mathbb{R}^{H\times W\times 3}$ with corresponding class labels $y_i \in \mathcal{C}$, where $\mathcal{C} = \{1, \ldots, C_n\}$ denotes the set of all possible classes. Therefore, labeled samples are typically represented as $\mathcal{X} = \{(x_i, y_i)\}_{i=1}^{N}$, where x_i is the image data and y_i is its corresponding ground-truth class label. The unlabeled samples are denoted as $\mathcal{U} = \{u_j\}_{j=1}^{M}$, where u_j represents an unlabeled image and M is the number of unlabeled samples.

The FixMatch framework comprises two main branches. In the first branch, weak augmentation (denoted as Aug_w) is applied to the labeled data, and supervised training is conducted using the provided class labels. In the second branch, for the unlabeled data, both strong augmentation (denoted as Aug_s) and weak augmentation are applied. The model first generates pseudo-labels for the weakly augmented samples. Subsequently, a confidence threshold τ is employed to filter out pseudo-labels with low confidence, effectively removing noisy labels. The remaining high-confidence pseudo-labels are used to compute the unsupervised loss based on the predictions of the strongly augmented samples.

This process leverages high-confidence pseudo-labels effectively, significantly enhancing the contribution of unlabeled data during training. The objective function is formalized as follows.

1) Supervised Loss L_x: The supervised loss is computed using the cross-entropy loss on the labeled data

$$L_x = \frac{1}{B}\sum_{b=1}^{B} \mathcal{H}\left(y_b, f_\theta\left(\text{Aug}_w\left(x_b\right)\right)\right) \tag{1}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:13:51 UTC from IEEE Xplore. Restrictions apply.

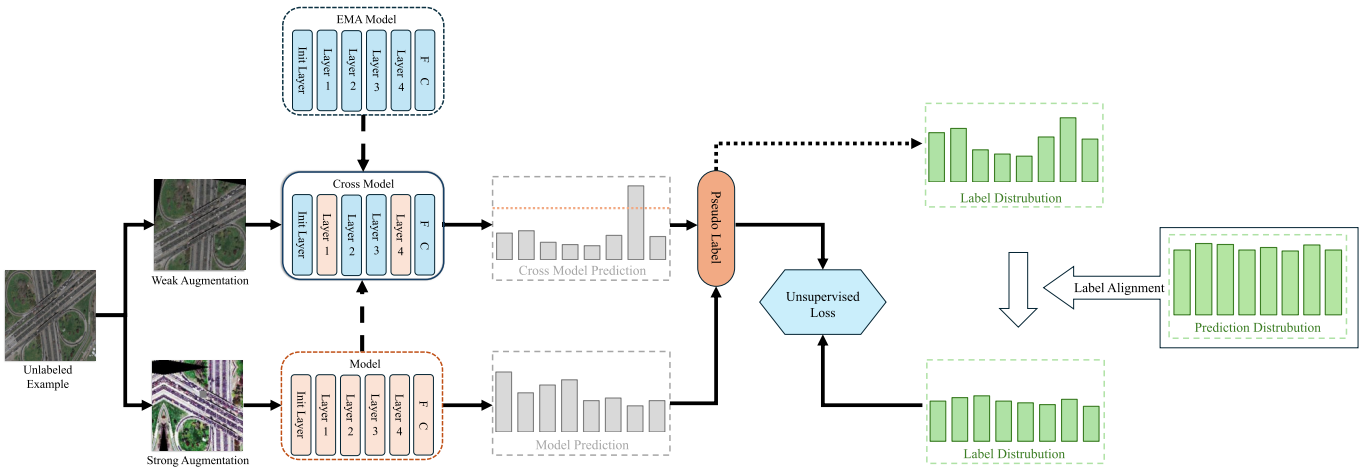


Fig. 1. Overview of the LCMatch framework, which consists of two core modules: layer cross and label alignment. First, layer cross is applied to randomly combine the model trained normally with the EMA model. Pseudo-labels are then selected based on a predefined threshold and adjusted according to the overall prediction results to ensure a more balanced distribution. Finally, after obtaining the unsupervised loss, an adaptive weighting mechanism is employed to adjust the weight of the unlabeled loss, further optimizing the model's performance.

where $f_\theta(\cdot)$ denotes the model's prediction function, B is the batch size of labeled data, and $\mathcal{H}(\cdot,\cdot)$ represents the cross-entropy loss.

2) Unsupervised Loss L_u: The unsupervised loss is defined based on the difference between the strongly augmented predictions and the pseudo-labels

$$L_u = \frac{1}{\mu B}\sum_{b=1}^{\mu B}\mathbb{I}\left(\max\left(p_b\right)\geq\tau\right)\cdot\mathcal{H}\left(q_b, f_\theta\left(\mathrm{Aug}_s\left(u_b\right)\right)\right) \tag{2}$$

where μ is the ratio of unlabeled to labeled data in a batch, $\mathbb{I}(\cdot)$ is the indicator function that filters pseudo-labels not meeting the confidence threshold τ, p_b is the predicted probability distribution for the weakly augmented sample, and $q_b = \arg\max(p_b)$ denotes the pseudo-label.

3) Total Loss Function: The combined loss in FixMatch is given by

$$L_{\text{fixmatch}} = L_x + \lambda L_u \tag{3}$$

where λ is a weighting coefficient that balances the contributions of the supervised and unsupervised losses.

B. Hierarchical Cross-Random Combination

To overcome the limitations of pseudo-labeling in the FixMatch framework, we propose the HCRC method. This approach improves the quality and robustness of pseudo-labels by integrating hierarchical feature fusion between the primary model and an EMA model, along with a stochastic perturbation mechanism. Specifically, in each training iteration, HCRC randomly selects convolutional layers from both the primary and EMA models to build a Cross Model for pseudo-label generation. Such random combinations introduce temporal knowledge fusion and structural variability, yielding pseudo-label predictions that are diverse, stable, and semantically expressive. The overall process is summarized in Algorithm 1.

The theoretical justification behind the effectiveness of HCRC primarily lies in two aspects. First, convolutional layers at different depths of a neural network encode inherently distinct features: lower layers typically capture detailed spatial structures such as edges and textures, whereas higher layers represent abstract semantic and global contextual information. Randomly combining features from these varying depths leverages the complementary strengths of spatial precision and semantic abstraction, effectively alleviating cognitive bias and enriching the representation capability of pseudo-labels. Second, the stochastic selection mechanism serves as a form of structural regularization akin to dropout. By continuously varying the combination of layers during training, HCRC prevents the model from becoming overly reliant on any specific feature extraction pathway. This structural perturbation significantly mitigates overfitting risks, enhancing model robustness and generalization capacity. We further substantiate these theoretical advantages through ablation studies detailed in Section IV, which confirm that removing the random combination mechanism or employing fixed-layer strategies results in a notable performance decline.

1) EMA Update Rule: Inspired by the Mean Teacher framework, the EMA model parameters are updated during training using the following rule:

$$\theta_{\text{EMA}}^{t+1} = \alpha\theta_{\text{EMA}}^{t} + (1-\alpha)\,\theta^{t} \tag{4}$$

where θ_{EMA}^{t} and θ^{t} represent the parameters of the EMA model and the primary model at iteration t, respectively, and α is the smoothing coefficient.

2) Random Combination Strategy: We introduce a random hierarchical combination strategy that generates a new Cross Model by randomly combining convolutional layers from both the EMA model and the primary model. The parameters of the Cross Model are defined as follows:

$$f^{\text{cross}} = \mathrm{Conv}_{1\times1}\left(\|_{l\in\mathcal{L}}\left(f_l^{p}\|f_l^{\text{EMA}}\right)\right). \tag{5}$$

where $\|$ denotes random selection. In our framework, we extract four intermediate-to-high-level feature maps from both the primary and EMA models. At each training iteration, a

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:13:51 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1 Layer-Wise Cross-Random Combination for Pseudo-Label Generation

Input: Feature layers from student and teacher networks:
$F = \{f_l^{(S)}, f_l^{(T)}\}$, where l is the layer index;
Layer indices used for combination: $\mathcal{L}$
Output: Composite feature representation $\hat{f}$ for pseudo-label prediction
Initialize $\hat{f} \leftarrow \emptyset$; // Initialize composite features
for $l \in \mathcal{L}$ **do**
  Randomly choose network $s_l \in \{S, T\}$;
  // Student or Teacher
  $\hat{f}_l \leftarrow f_l^{(s_l)}$; // Select the layer output from the chosen network
  Concatenate $\hat{f}_l$ into composite representation
  $\hat{f} \leftarrow \hat{f} \cup \hat{f}_l$
Use the composite features $\hat{f}$ to predict pseudo-labels $\hat{z} = \text{Classifier}(\hat{f})$
return $\hat{z}$; // Return pseudo-label prediction

random subset of layers $\mathcal{L} \subset \{2, 3, 4, 5\}$ is selected, and the corresponding feature maps from both models are concatenated along the channel dimension. To standardize output dimensions and enhance representational capacity, the fused feature map is passed through a 1×1 convolutional layer to produce the combined representation f^{cross}, which is then used for pseudo-label generation by the Cross Model. This randomized layer combination strategy introduces structural diversity, reinforces semantic complementarity, and acts as an implicit regularizer, thereby improving the stability and generalization of pseudo-labels.

Using the Cross Model, pseudo-labels are generated for weakly augmented images as follows:

$$\hat{y}_{\text{cross}} = f_{\theta_{\text{cross}}}\left(\text{Aug}_w(x)\right) \tag{6}$$

where $\hat{y}_{\text{cross}}$ denotes the pseudo-labels generated by the Cross Model, $f_{\theta_{\text{cross}}}$ is the classification function of the Cross Model, and $\text{Aug}_w(x)$ is the weakly augmented image sample.

a) Threshold selection for $\hat{y}_{cross}$: To ensure that only high-confidence pseudo-labels are utilized, we apply the confidence threshold τ as follows:

$$\hat{y}_{\text{cross}} = \begin{cases} f_{\theta_{\text{cross}}}\left(\text{Aug}_w(x)\right), & \text{if } \max\left(f_{\theta_{\text{cross}}}\left(\text{Aug}_w(x)\right)\right) \geq \tau \\ \text{discard}, & \text{otherwise.} \end{cases} \tag{7}$$

By introducing randomness and feature fusion via the Cross Model, the HCRC method enhances the diversity and robustness of pseudo-label predictions. This significantly improves model stability, reduces the negative impact of erroneous pseudo-labels during training, and enhances overall model accuracy and robustness.

C. Adaptive Weighting Mechanism

As training progresses, the incremental information provided by labeled data gradually diminishes as the model's understanding of the labeled data stabilizes. To address this issue, we propose a dynamic weighting function that better balances the contributions of labeled and unlabeled data during training. This function incorporates two key components: the *mask ratio* (m) and the *training progress* ((t/T)).

The mask ratio represents the current learning status of the model with respect to the labeled data, reflecting the extent to which the labeled data has been effectively utilized. The training progress is defined as the normalized value of the training epoch, capturing the cumulative effect of training time. To emphasize the importance of deep feature learning in the later stages of training, we introduce a quadratic term into the training process to achieve nonlinear growth of the dynamic weight.

The dynamic weight is computed as follows:

$$d = d_{\text{init}} \cdot \left(m + \left(\frac{t}{T}\right)^2\right) \tag{8}$$

where d_{init} is the initial weight, t is the current training epoch, and T is the total number of epochs.

Here, m denotes the mask ratio, which quantifies the proportion of high-confidence pseudo-labels in the current batch of unlabeled data. Specifically, it is calculated as follows:

$$m = \frac{1}{N}\sum_{i=1}^{N} \mathbb{I}\left(\max\left(\hat{y}_i\right) \geq \tau\right) \tag{9}$$

where N is the number of unlabeled samples, $\hat{y}_i$ is the predicted class probability vector for sample i, and τ is a fixed confidence threshold. The indicator function $\mathbb{I}(\cdot)$ returns 1 if the condition is satisfied and 0 otherwise.

The mask ratio m reflects the density of reliable pseudo-labels at the current training step. The quadratic term

$$\left(\frac{t}{T}\right)^2$$

introduces a nonlinear growth component to emphasize supervised learning at later epochs. Together, this formulation ensures that the dynamic weight d adaptively balances between learning progress and pseudo-label reliability.

This adaptive weighting mechanism enables the model to effectively leverage labeled data while dynamically adjusting the influence of unlabeled data, thereby enhancing training robustness and final performance.

D. Label Alignment

To address class imbalance in pseudo-labels during semi-supervised training, we propose an adaptive label alignment mechanism. This mechanism dynamically adjusts the class-wise loss contributions according to the pseudo-label distribution, ensuring that underrepresented categories receive sufficient gradient signal, especially during the early training stages when pseudo-labels are often biased toward dominant classes. The overall process is summarized in Algorithm 2.

We begin by estimating the relative frequency of each class c based on current pseudo-label predictions

$$p_c = \frac{1}{N}\sum_{i=1}^{N} \hat{y}_{i,c} \tag{10}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:13:51 UTC from IEEE Xplore. Restrictions apply.

Algorithm 2 Label Alignment for Pseudo-Label Refinement

Input: Predicted pseudo-label distribution: $\hat{P} \in \mathbb{R}^{N\times C}$;
Target class prior (or smoothed distribution): $\tilde{P} \in \mathbb{R}^{C}$;
Smoothing factor: α
Output: Aligned pseudo-labels: $\hat{Y}_{aligned}$
// Step 1: Compute current pseudo-label histogram
$H = \frac{1}{N}\sum_{i=1}^{N} \hat{P}_i$
// Step 2: Adjust with prior distribution or smoothing
$w = \frac{\tilde{P}}{H+\epsilon}$; // Re-weighting vector
$w \leftarrow \text{Normalize}(w)$; // Ensure $\sum w = C$
// Step 3: Apply label alignment to pseudo-labels
for $i = 1$ **to** N **do**
  $\hat{P}_i^{aligned} = \text{Softmax}(\log(\hat{P}_i)\cdot\alpha)\cdot w$
// Step 4: Get final aligned hard labels
$\hat{Y}_{aligned} = \arg\max(\hat{P}_i^{aligned})$
return $\hat{Y}_{aligned}$

where $\hat{y}_{i,c} \in [0, 1]$ is the predicted probability (after Softmax) of sample i belonging to class c and N is the total number of unlabeled samples in the batch.

Next, we define an unnormalized class-wise weight that inversely scales with the frequency p_c. This ensures that rare classes receive larger weights

$$w_c = \frac{\alpha_t}{p_c + \epsilon}, \quad \text{with} \quad \alpha_t = e^{-\lambda\cdot\frac{t}{T}} \tag{11}$$

where ϵ prevents numerical instability, t is the current epoch, T is the total number of training epochs, and λ is the decay coefficient. The exponential decay factor α_t gradually reduces the strength of label alignment over time, giving the model more autonomy in later stages.

To make the weights comparable across classes, we apply normalization

$$\hat{w}_c = \frac{w_c}{\sum_{c'=1}^{C} w_{c'}} \tag{12}$$

where C is the number of classes. This normalization ensures that the overall loss magnitude remains stable.

Finally, the normalized class-wise weights are incorporated into the unsupervised learning objective using a weighted cross-entropy loss

$$L_{\text{unlabel}} = -\sum_{i=1}^{N}\sum_{c=1}^{C} \hat{w}_c \cdot \tilde{y}_{i,c} \cdot \log\left(\hat{y}_{i,c}\right) \tag{13}$$

where $\tilde{y}_{i,c} \in [0, 1]$ denotes the soft pseudo-label for sample i and class c, typically obtained via sharpening or confidence thresholding. These soft labels reflect prediction uncertainty, enabling the model to make informed use of high-confidence samples while down-weighting ambiguous ones.

In summary, the proposed label alignment mechanism stabilizes training under class imbalance and pseudo-label noise by reweighting supervision signals based on real-time pseudo-label statistics. This contributes to more robust and balanced learning across all categories.

E. Overall Loss Function

In our proposed LCMatch framework, the loss function builds upon FixMatch by incorporating the adaptive weighting and label alignment strategies. The loss is divided into two components: the supervised loss and the unsupervised loss. For the supervised loss, our method adopts the same cross-entropy loss as FixMatch

$$L_x = \frac{1}{B}\sum_{b=1}^{B} \mathcal{H}\left(y_b, f_\theta\left(\text{Aug}_w\left(x_b\right)\right)\right). \tag{14}$$

For the unsupervised loss, as described in Section III-D, it is derived based on cross-layer to get pseudo-labels and consistency learning. Furthermore, by incorporating the adaptive weighting strategy presented in Section III-C, the overall loss function is expressed as follows:

$$L = L_x + d \cdot L_{\text{unlabel}} \tag{15}$$

where d represents the adaptive weight dynamically calculated as described in Section III-C and L_{unlabel} is formulated in Section III-D. This formulation ensures that the model effectively leverages labeled data for supervised learning while adaptively incorporating unlabeled data through the proposed methods, thereby enhancing overall training performance.

F. Comparison With Existing Methods

LCMatch is designed to address key limitations of existing SSL methods, such as FixMatch and Mean Teacher, by introducing three novel components: HCRC, adaptive weighting, and label alignment. Compared to FixMatch, which generates pseudo-labels based on the predictions of a single model with weak augmentation, LCMatch leverages a Cross Model constructed from hierarchical feature combinations of both the primary model and its EMA counterpart. This randomized and multilevel fusion introduces temporal diversity and structural variability, significantly improving the quality and semantic expressiveness of pseudo-labels while mitigating confirmation bias.

Unlike Mean Teacher, which primarily uses the EMA model for consistency regularization, LCMatch deeply integrates the EMA model into the pseudo-labeling process. Moreover, LCMatch introduces an adaptive weighting mechanism that dynamically balances supervised and unsupervised loss contributions based on training progress and pseudo-label reliability. The label alignment module further compensates for class imbalance in pseudo-labels by reweighting the loss according to real-time category distributions.

Taken together, these innovations distinguish LCMatch from existing frameworks by improving pseudo-label robustness, enhancing training stability, and enabling more balanced learning across diverse categories, particularly in remote sensing scenarios characterized by class imbalance and interclass visual similarity.

IV. Experiments

In this section, we conduct experiments on three widely used remote sensing scene classification datasets to assess the performance of the proposed method. First, we provide a comprehensive description of the datasets and the experimental setup. Next, we compare the proposed method against state-of-the-art SSL approaches. Finally, we present ablation studies to examine the contributions and effectiveness of each module in the proposed framework.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:13:51 UTC from IEEE Xplore. Restrictions apply.

A. Datasets

1) AID [55]: The AID dataset is a large-scale benchmark designed specifically for remote sensing scene classification, with images sourced from Google Earth. It consists of 10 000 high-resolution RGB images, spanning 30 categories such as meadows, snowy terrains, and residential areas, which capture significant variations in environmental and scene characteristics. Each image has a fixed resolution of 600 × 600 pixels, ensuring uniform spatial quality and comparability across the dataset. The number of samples per category ranges from 200 to 400, providing sufficient data for robust model training and performance evaluation. Due to its diversity and broad coverage of scene types, the AID dataset is a pivotal benchmark in the remote sensing image classification field.

2) UCMerced [56]: The UC Merced Land Use dataset, developed by the University of California, Merced, is a widely recognized benchmark designed for land use analysis and classification tasks. It contains 2100 high-resolution RGB images across 21 distinct categories, including farmlands, airports, residential areas, and commercial zones. Each category consists of exactly 100 images, ensuring a balanced distribution across classes. The images have a resolution of 256 × 256 pixels, with a spatial resolution of 0.3 m per pixel, derived from aerial imagery. This dataset provides a comprehensive representation of various land use patterns and is extensively used in remote sensing research.

3) NWPU-RESISC45 [57]: The NWPU-RESISC45 dataset is a comprehensive benchmark designed for remote sensing scene classification, comprising 31 500 RGB images across 45 categories. These categories encompass a broad spectrum of scenes, including beaches, bridges, and wind turbines, effectively reflecting diverse geographic regions and land-use types. Each image is uniformly sized at 256 × 256 pixels, with exactly 700 samples per category, ensuring balanced distribution and representativeness of the dataset.

B. Implementation Details

1) Dataset Settings: In our experiments, we followed the dataset splitting strategies outlined in the respective original papers. Specifically, the UC Merced dataset was split into training and testing sets with an 80:20 ratio, the AID dataset with a 50:50 ratio, and the NWPU-RESISC45 dataset with a 20:80 ratio. To evaluate the performance of our method under varying amounts of labeled data, we selected 3, 5, and 10 samples per class from the training set to construct labeled subsets of different sizes, while the remaining data in the training set were used as unlabeled subsets. This setup facilitates a comprehensive evaluation of our method's effectiveness in SSL scenarios with limited labeled data.

2) Training Settings: We use the ResNet50 [41] model as the feature extractor, initializing it with pretrained weights from ImageNet-1K. Dimensionality reduction and dropout are applied in the final fully connected layer before producing the prediction results. The EMA model is configured identically to the training model, with an EMA decay (EMA_decay) set to 0.99. We train the model using the SGD optimizer with a weight decay of 0.001 and a momentum of 0.9. The training process spans 400 epochs, with a batch size of 8 for labeled data and 56 for unlabeled data. During the first 20 epochs (warmup phase), the learning rate linearly increases to 0.001, after which it decays according to a cosine scheduler. The temperature parameter and confidence threshold τ for pseudo-label generation are set to 1 and 0.95, respectively. To reproduce the baseline experiments, we utilize the unified SSL codebase (USB) [58].

For data augmentation, we employ random horizontal flipping and random affine transformations as weak augmentation techniques. Additionally, we use RandAugment as a strong augmentation method, extending the weak augmentation strategy. All experiments are conducted on a server equipped with a single NVIDIA RTX 3090 GPU, and the entire framework is implemented using the PyTorch 1.12 library. The evaluation metric is overall accuracy (OA), with the mean and standard deviation of accuracy reported based on three replicate experiments.

C. Experimental Results

In accordance with the aforementioned experimental protocols, our method is systematically compared against state-of-the-art semi-supervised classification approaches. We evaluate and report scene classification accuracies under varying numbers of labeled samples across the three aforementioned datasets. Table I provides a comparative analysis of the proposed LCMatch method against leading semi-supervised classification approaches on the UCMerced, AID, and NWPU-RESISC45 datasets. Furthermore, Figs. 2–4 depict the confusion matrices for UCMerced, AID, and NWPU-RESISC45 under the 10-labeled-samples-per-class setting, offering deeper insights into classification performance.

1) UCMerced Dataset: The experimental results on the UC Merced dataset, presented in the first column of Table I, highlight the significant advantages of LCMatch over mainstream semi-supervised classification methods. Specifically, LCMatch achieves outstanding classification performance with accuracies of 85.63%, 96.10%, and 98.10% when utilizing 3, 5, and 10 labeled samples per class, respectively.

A key observation is that LCMatch exhibits superior performance, particularly when a larger number of labeled samples are available. When the number of labeled samples per class reaches 5 or 10, LCMatch surpasses the second-best method by 4.43% and 2.31%, respectively. Furthermore, with 10 labeled samples per class, its performance closely approximates that of the fully supervised model, further validating its effectiveness.

To provide a more detailed analysis of class-wise classification performance, Fig. 2 presents the confusion matrix of LCMatch under the setting of 10 labeled samples per class. Despite the limited number of labeled training samples, LCMatch achieves 100% accuracy in most categories, including "tenniscourt," "golfcourse," "airplane," "baseballdiamond," and "beach." However, classification performance is suboptimal for the "sparse residential" category, which is occasionally confused with "golfcourse," and the "river" category, which is partially misclassified as "beach."

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:13:51 UTC from IEEE Xplore. Restrictions apply.

TABLE I
PERFORMANCE COMPARISON ON UCM, AID, AND NWPU-RESISC45 DATASETS

Dataset		UCM			AID			NWPU-RESISC45		
Method	**Publication**	**3 Labels**	**5 Labels**	**10 Labels**	**3 Labels**	**5 Labels**	**10 Labels**	**3 Labels**	**5 Labels**	**10 Labels**
Full Label	-		99.28			96.34			92.71	
Full Supervised	-	51.11±0.60	61.82±4.97	91.35±0.37	47.73±2.14	59.18±1.22	83.54±0.66	39.27±2.05	49.78±1.43	61.62±1.64
Π-Model [47]	ICLR'17	71.19±1.56	86.11±0.60	92.06±0.84	62.35±1.18	75.86±2.17	84.94±0.91	53.83±1.09	62.71±2.13	75.37±0.64
Mean Teacher [48]	NeurIPS'17	52.74±0.50	85.88±0.27	89.53±0.86	63.83±2.28	76.47±1.35	85.79±1.61	39.97±1.23	50.04±3.41	76.00±0.26
UDA [50]	NeurIPS'20	59.37±3.47	82.70±1.72	90.56±0.97	70.76±1.98	83.59±0.06	88.87±0.92	54.39±1.82	69.34±1.47	80.47±0.18
FixMatch [31]	NeurIPS'20	60.23±4.43	84.77±1.01	91.49±0.92	71.64±2.40	80.64±1.56	88.00±0.25	57.57±1.41	68.78±0.96	81.69±1.11
FlexMatch [52]	NeurIPS'21	59.96±5.82	87.86±0.86	93.49±0.14	66.16±4.85	81.37±1.24	88.00±0.72	67.43±1.95	68.75±1.14	80.15±0.22
SimMatch [59]	CVPR'22	80.79±1.44	91.67±0.41	95.79±0.73	70.15±1.92	88.29±0.38	88.68±0.67	58.00±0.69	78.75±3.50	83.29±0.79
FreeMatch [60]	ICLR'23	67.14±2.06	87.70±1.46	92.86±2.90	**79.68±2.01**	83.83±0.80	87.44±0.34	66.84±1.20	76.65±1.15	84.59±0.10
SoftMatch [61]	ICLR'23	63.57±1.33	91.27±0.73	92.94±1.44	78.93±0.31	85.97±1.46	89.59±0.27	66.52±1.34	73.05±1.34	84.11±0.28
EPASS [62]	WACV'24	79.05±0.55	91.43±0.83	95.32±1.22	77.47±2.14	87.79±0.34	90.34±0.86	68.96±0.52	75.97±1.23	80.8±1.19
Ours	-	**85.63±0.55**	**96.10±0.24**	**98.10±0.24**	75.99±1.05	**88.91±0.46**	**92.13±0.25**	**74.47±0.47**	**82.46±0.06**	**86.67±0.18**

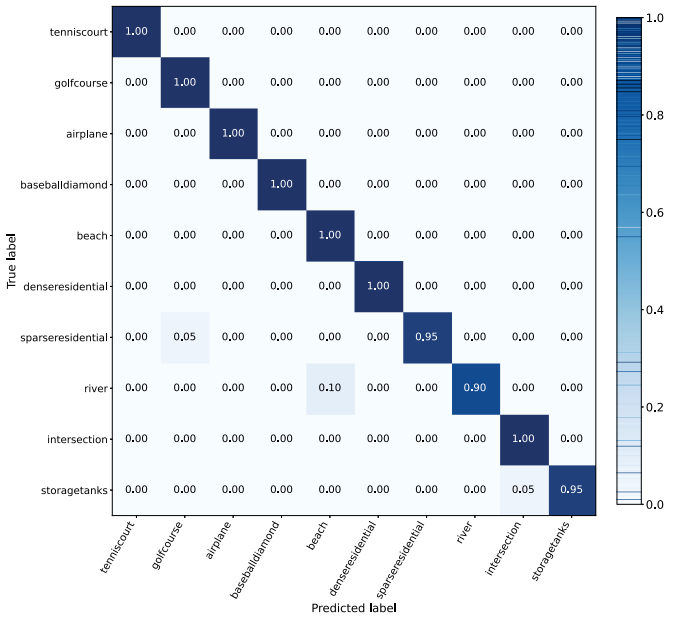


Fig. 2. Confusion matrix of selected challenging and representative classes obtained by LCMatch on the UCM dataset with ten labeled samples per class.

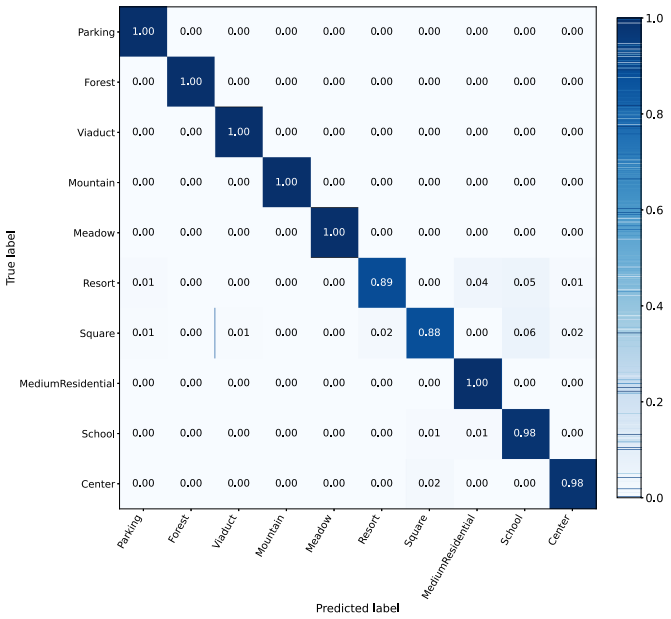


Fig. 3. Confusion matrix of selected challenging and representative classes obtained by LCMatch on the AID dataset with ten labeled samples per class.

These results indicate that while LCMatch performs well on most classes, visually similar land-use categories remain challenging under low-label conditions.

This misclassification is primarily attributed to the high visual similarity among certain categories, which increases classification difficulty. Overall, the experimental results confirm that even with a limited number of labeled training samples, LCMatch can achieve robust and high-performance classification on the UC Merced dataset.

2) AID Dataset: The second column of Table I presents the performance of LCMatch on the AID dataset. Experimental results demonstrate that LCMatch significantly outperforms existing methods, particularly when more labeled data is available. Specifically, when using only 3 labeled samples per class, LCMatch achieves an accuracy of 75.99%, slightly lower than the previously best-performing method, FreeMatch. As the number of labeled samples per class increases to 5, the accuracy of LCMatch improves to 88.91%, surpassing the second-best method, SimMatch, by 0.62%. When using 10 labeled samples per class, LCMatch achieves an accuracy of 90%, outperforming the previously best method, EPASS, by 1.79%.

Fig. 3 presents the partial confusion matrix of LCMatch on the AID dataset with 10 labeled samples per class. It can be observed that LCMatch achieves perfect classification in five categories: "Parking," "Forest," "Viaduct," "Mountain," and "Meadow." However, notable misclassifications occur in several categories, such as "Resort" misclassified as "Medium Residential" and "School," and "Square" misclassified as "School" and "Center."

These encouraging results indicate that LCMatch effectively classifies categories in the AID dataset, achieving high classification performance even under conditions of limited labeled samples.

3) NWPU Dataset: As presented in the third column of Table I, LCMatch demonstrates superior performance

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:13:51 UTC from IEEE Xplore. Restrictions apply.

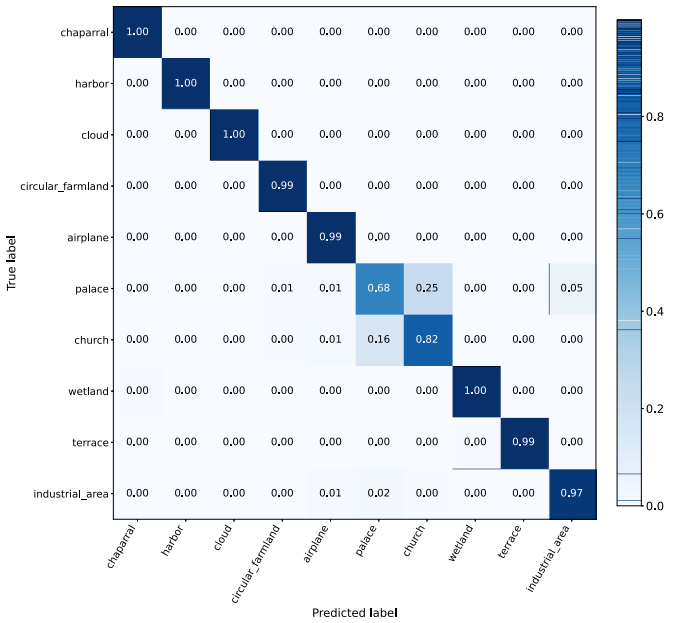


Fig. 4. Confusion matrix of selected challenging and representative classes obtained by LCMatch on the NWPU-RESISC45 dataset with ten labeled samples per class.

on the NWPU-RESISC45 dataset. Specifically, with 3, 5, and 10 labeled samples per class, LCMatch achieves classification accuracies of 74.47%, 82.46%, and 86.67%, respectively.

Compared to state-of-the-art methods, LCMatch exhibits notable improvements. Under the 3-label setting, our method, LCMatch, outperforms the current best method, EPASS, by 5.51%. When the number of labeled samples increases to 5 per class, LCMatch surpasses SimMatch by 3.71%. Furthermore, with 10 labeled samples per class, LCMatch outperforms the previously best-performing method, FreeMatch, by 2.08% in classification accuracy. Additionally, LCMatch maintains a low standard deviation across different settings, particularly when 5 labeled samples per class are available, where the standard deviation is only 0.06%. This observation indicates that LCMatch is highly stable and robust under limited labeled data conditions.

Fig. 4 presents the confusion matrix of LCMatch on a subset of the NWPU-RESISC45 dataset, using 10 labeled samples per class. It can be observed that LCMatch achieves perfect or near-perfect accuracy in several categories, including "chaparral," "circular farmland," "cloud," "wetland," and "industrial area." These results demonstrate the model's capability to confidently classify visually distinctive scenes.

However, classification performance is notably lower for more visually similar categories. In particular, the "palace" class is often confused with "church" and "industrial area," resulting in an accuracy of 68%. Similarly, the "church" category shows some confusion with "palace," indicating mutual semantic overlap between these two classes. These confusions are likely due to architectural similarities and shared contextual features such as surrounding vegetation or urban layout.

These experimental results further validate the effectiveness of LCMatch on the NWPU-RESISC45 dataset, demonstrating that even with limited labeled data, LCMatch maintains high classification accuracy and stability.

4) Pseudo-Label Usage and Accuracy: As shown in Fig. 5, we compare the pseudo-label utilization and corresponding accuracy of our method, Mean Teacher, and FixMatch on the UCM dataset with 10 labeled samples. It is observed that the utilization rate of our method increases more gradually compared to FixMatch and Mean Teacher. However, our method demonstrates relatively more stable accuracy over time, with a smaller decline. This behavior can be attributed to the label alignment technique we proposed, which effectively preserves the model's performance and mitigates the decrease in accuracy over time.

5) Analysis of Failure Cases: To better understand the limitations of LCMatch, we examine the confusion matrices and identify failure cases across three benchmark datasets: NWPU-RESISC45, UCM, and AID. For each dataset, we select two categories with relatively low classification accuracy and analyze the potential causes of misclassification.

a) NWPU-RESISC45: As discussed previously, the "palace" category achieves the lowest accuracy, primarily due to its visual similarity with "commercial areas" and "churches" in terms of architectural features, spatial layout, and surroundings. Another underperforming class is "railway station," which is often misclassified as "freeway" or "airport." This may result from overlapping structural components such as rail tracks, large open platforms, and nearby roads, all of which appear visually similar from overhead views.

b) UCM: The "dense residential" class is frequently confused with "medium residential" or "mobile home park" due to similar roof patterns, building density, and urban textures (not shown in Fig. 2). The fine-grained boundary between different residential types makes accurate classification difficult. Additionally, the "storage tanks" class is often mistaken for "roundabout" or "baseball diamond," as all contain circular geometric features with similar spectral profiles, particularly in RGB imagery.

c) AID: In the AID dataset, "School" is commonly confused with "Park" or "Stadium" because of shared green spaces, open areas, and building clusters, especially when the school lacks distinguishable shapes. Another notable failure case is "Bridge," which is frequently misclassified as "River" or "Highway." This is likely because bridges are small relative to the entire image and often blend into the background, making them difficult to detect with semi-supervision.

These observations suggest that most failure cases stem from high interclass visual similarity, ambiguous boundaries, or small object sizes within the image. To address these challenges, future improvements may include incorporating multiscale feature extraction, leveraging additional modalities,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:13:51 UTC from IEEE Xplore. Restrictions apply.

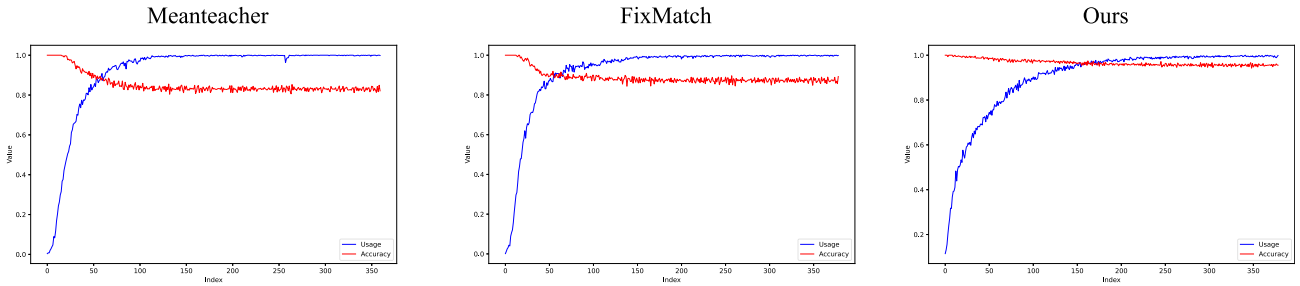


Fig. 5. Comparison of pseudo-label utilization and accuracy among our method, Mean Teacher, and FixMatch.

TABLE II
ABLATION STUDY ON THE NWPU DATASET WITH 10 LABELED SAMPLES PER CLASS

HCRC	Label Alignment	Adaptive Weighting	Accuracy (%)
			88.28
✓			91.76
✓	✓		91.92
✓	✓	✓	**92.13**

or introducing spatial attention mechanisms tailored for hard-to-distinguish categories.

D. Ablation Experiment

In this section, we perform ablation studies on the AID dataset to identify the optimal parameter configurations. This is crucial for ensuring both the representativeness and reliability of the experimental results. All models are trained for 400 epochs to guarantee a fair comparison, allowing for a comprehensive evaluation of the impact of different parameters on model performance.

1) Functional Differences of Model Components: As shown in Table II, we conduct ablation experiments on the NWPU dataset with 10 labeled samples per class to evaluate the effectiveness of each proposed component. Introducing the HCRC module alone improves accuracy from 88.28% to 91.76%, demonstrating its capability to generate more robust pseudo-labels. Further incorporating the label alignment mechanism boosts performance to 91.92%, while integrating all three components—including the adaptive weighting strategy—leads to the best result of 92.13%. These results confirm the complementary benefits of our hierarchical and adaptive design.

2) Impact of Confidence Threshold τ: The choice of confidence threshold τ is a pivotal factor in SSL, as it controls the tradeoff between the quantity and quality of pseudo-labels. To assess the effect of τ on model performance, we conducted experiments with both 5-label and 10-label settings, varying τ values across 0.9, 0.95, 0.98, and 1. The corresponding results are presented in Fig. 6.

As shown in the figure, increasing τ consistently enhances classification accuracy. This improvement can be attributed to the effective filtering of low-confidence pseudo-labels, which reduces the impact of incorrect predictions. The model achieves its best performance when τ is set to 0.95. However,

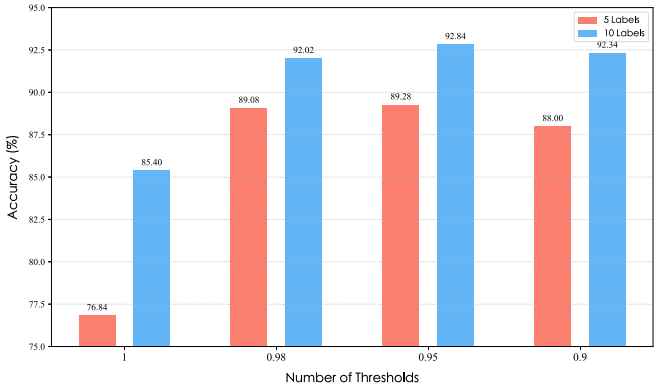


Fig. 6. Performance comparison for different confidence thresholds.

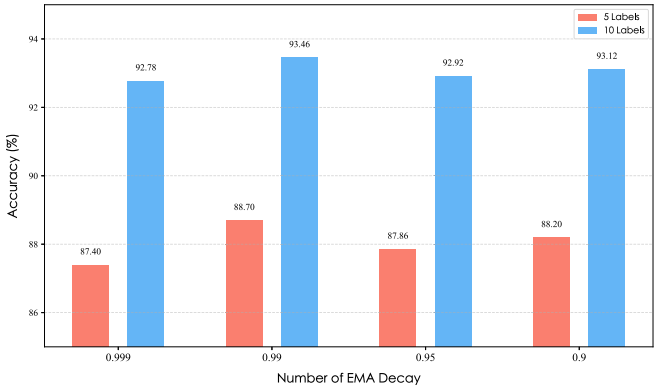


Fig. 7. Performance comparison for different EMA decay rates.

further increasing the threshold leads to a noticeable drop in accuracy. This decline occurs because a higher confidence threshold limits the number of samples that are assigned pseudo-labels, leading to underutilization of the unlabeled data. As a result, the model's generalization ability is impaired, and overall performance suffers.

3) Effect of EMA Decay: The LCMatch framework incorporates the EMA model, making the selection of the EMA decay parameter a critical factor in ensuring both model stability and performance. To investigate the impact of EMA decay on classification accuracy, we performed experiments using four different decay rates: 0.999, 0.99, 0.95, and 0.9, under identical training conditions. The results are illustrated in Fig. 7.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:13:51 UTC from IEEE Xplore. Restrictions apply.

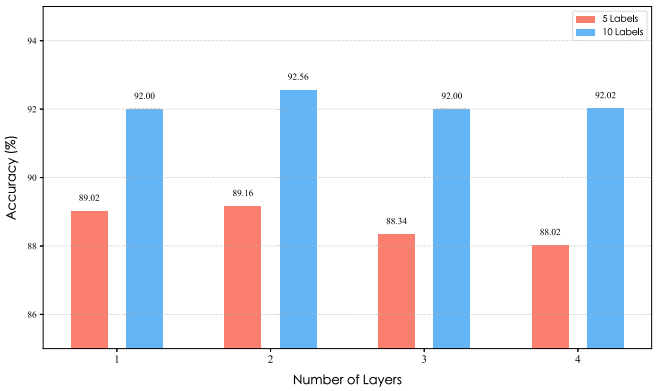


Fig. 8. Performance comparison for different numbers of updated layers.

TABLE III
COMPARISON OF COMPUTATIONAL COST AMONG METHODS

Method	FLOPs	Params	Epoch Timing (s)
FixMatch	4.132G	24.04M	5.18
Mean Teacher	4.132G	24.04M	5.33
Ours (LCMatch)	4.132G	24.04M	5.41

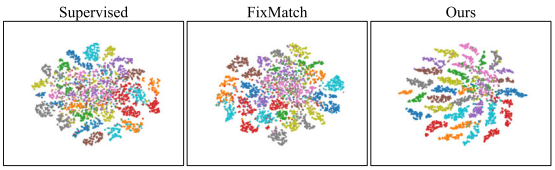


Fig. 9. t-SNE visualization of high-dimensional features learned by supervised learning, FixMatch, and the proposed LCMatch.

From the experimental results, it is evident that the highest classification performance is achieved when the EMA decay is set to 0.99. This finding suggests that a well-tuned EMA update strategy strikes an effective balance between model stability and adaptability, enhancing overall performance. Specifically, when the EMA decay is too large, the EMA model updates too slowly, reducing its responsiveness to the main model, which limits its adaptability. Conversely, when the EMA decay is too small, the EMA model updates too rapidly, introducing excessive noise that may hinder convergence and degrade performance.

These results emphasize the importance of carefully selecting the EMA decay parameter in optimizing pseudo-label learning within LCMatch, providing empirical support for hyperparameter tuning in EMA-based SSL frameworks.

4) Effect of Number of Layers: The LCMatch framework employs the EMA model, where information is transferred from the main model to the EMA model, and pseudo-labels are updated by a random hierarchical combination of layers from both models. Therefore, the selection of the number of layers to be updated is a crucial factor influencing model performance. The results of this experiment are presented in Fig. 8.

To investigate the impact of updating different hierarchical layers on performance, we adopt a layer combination strategy, in which a random selection of the final layers from the main model is updated. The number of updated layers is varied to 1, 2, 3, and 4, respectively, to explore how different hierarchical update strategies affect classification accuracy.

The experimental results show that the model achieves optimal performance when 2 layers are randomly selected for updating. This suggests that a moderate level of hierarchical updates enhances the quality of pseudo-labels and improves the model's robustness. However, selecting too few layers limits the model's generalization capacity, preventing full utilization of the EMA model's learning potential. Conversely, updating too many layers reduces the stability of the EMA model, potentially introducing excessive noise that undermines classification accuracy.

These findings further highlight the importance of a well-designed hierarchical update strategy in optimizing pseudo-labels within LCMatch, offering theoretical guidance for determining the appropriate number of layers to update.

E. Efficiency Analysis and Computational Cost

Table III compares the computational cost and training efficiency of three representative SSL methods: FixMatch, Mean Teacher, and our proposed LCMatch, all implemented with the same backbone (ResNet-50). As observed, the three methods share identical model complexity in terms of FLOPs (4.132G) and parameter count (24.04M), indicating that our method does not introduce additional architectural overhead.

Although LCMatch integrates additional mechanisms such as Cross Model feature fusion and adaptive label alignment, its average training time per epoch is only marginally higher (5.41 s) than that of FixMatch (5.18 s) and Mean Teacher (5.33 s). This demonstrates that LCMatch maintains high computational efficiency while achieving enhanced learning performance. The result confirms the practicality and scalability of our method, offering a favorable tradeoff between performance gains and training cost.

F. Visualization Results

1) Visualization of Deep Feature Representation: Fig. 9 presents a t-SNE [63] visualization of high-dimensional feature representations, reduced to two dimensions, for the AID validation set. The results indicate that the feature distribution learned by supervised methods, which rely solely on labeled data, exhibits considerable overlap between different classes, making class separation challenging. In contrast, both FixMatch and our proposed LCMatch method produce feature distributions with significantly reduced intraclass distances, resulting in a clearer distinction between classes. Notably, compared to FixMatch, LCMatch yields fewer scattered points, demonstrating that the use of multilevel pseudo-labels in LCMatch facilitates the learning of more discriminative features, thereby improving the model's ability to separate categories effectively.

2) Visualization of Prediction Results: Class activation mapping (CAM) [64] is a popular technique for visualizing

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:13:51 UTC from IEEE Xplore. Restrictions apply.

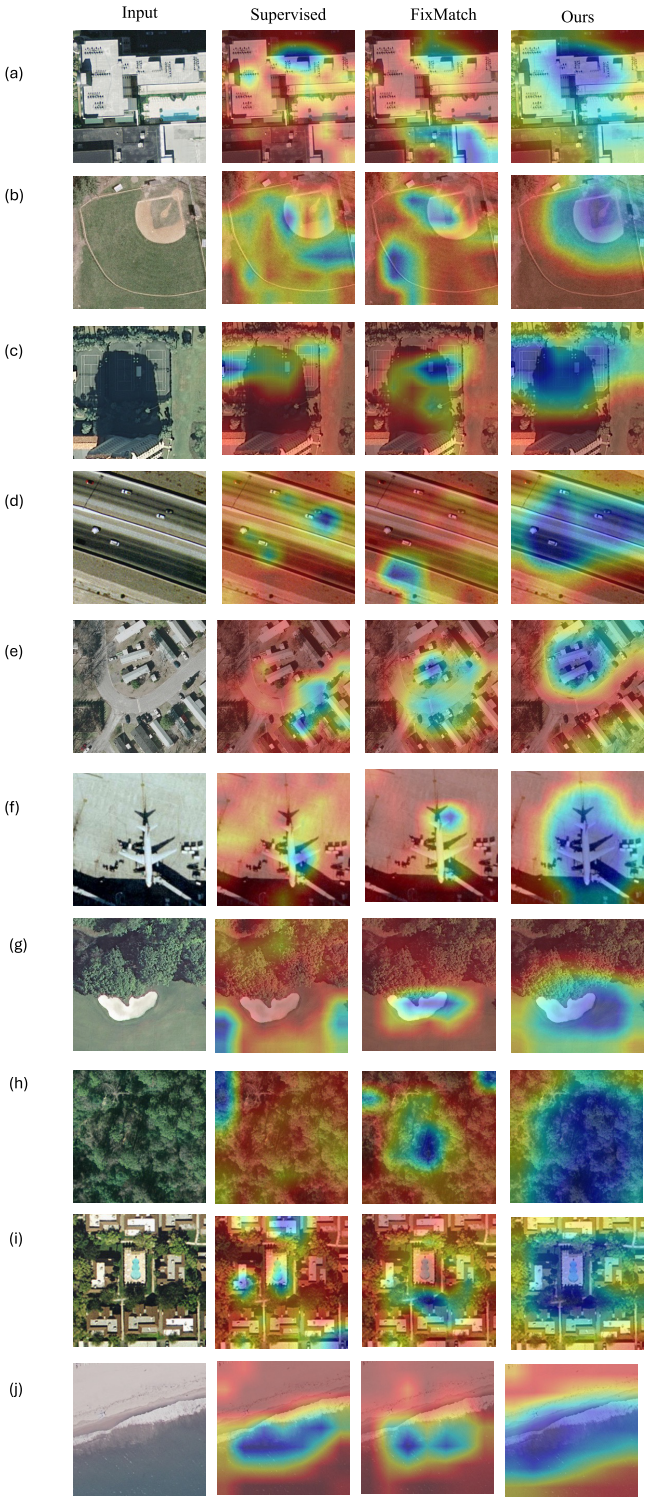


Fig. 10. Class activation maps for different scene types generated by the supervised method, FixMatch, and the proposed LCMatch. (a) Buildings. (b) Baseball diamond. (c) Tennis court. (d) Freeway. (e) Mobile home park. (f) Airplane. (g) Golfcourse. (h) Forest. (i) Denseresidential. (j) Beach.

model predictions by highlighting regions of the image that contribute most to the final output. In Fig. 10, we apply the CAM method to generate response maps for various scene images from the UCM dataset's validation set. In these maps, regions with more intense blue indicate areas of higher importance to the model's predictions.

Compared to traditional supervised learning methods and FixMatch, LCMatch demonstrates a superior ability to focus on key object regions, especially in complex backgrounds. This enables LCMatch to more accurately recognize and localize target objects. The enhanced spatial attention mechanism not only improves the model's prediction accuracy but also bolsters its robustness, particularly when dealing with diverse and challenging scenes. These results vividly highlight the advantages of LCMatch in handling a variety of scene types and object categories with greater precision.

3) Visualization of Failure Cases and Limitations: To complement the quantitative results, we further perform a qualitative analysis using Grad-CAM visualizations to investigate the potential limitations of our method. As shown in Fig. 11, these visualizations highlight the attention regions during inference under various challenging scenarios.

In the *tennis court* example, the model misclassifies the image due to attending to irrelevant background areas such as parking lots and shadows, indicating an over-reliance on local textures. The *runway* sample shows overly narrow focus on linear markings, ignoring the full semantic layout, suggesting insufficient use of global contextual cues. In the *river* and *golf course* scenes, the attention regions are scattered or misaligned, likely caused by error accumulation from pseudo-label noise. The *overpass* case shows distributed attention across multiple roads, highlighting the model's confusion in structurally complex scenes.

These examples reveal model limitations such as overfitting to local cues, propagation of pseudo-label errors, and decreased robustness in ambiguous contexts. Future work may explore uncertainty-aware attention correction and adaptive label refinement strategies to mitigate these issues.

G. Statistical Significance Evaluation

To further investigate the robustness of our method, we conducted statistical significance testing across 9 experimental configurations using 11 SSL methods. The Friedman test was first employed to determine whether significant performance differences exist. The test yielded a statistic $Q = 69.54$ with $p = 5.45 \times 10^{-11}$, confirming statistically significant variation among the methods.

Subsequently, the Bonferroni–Dunn post hoc test was applied, using our method as the control. The critical difference (CD) was computed as 4.39. As shown in Table IV, methods such as EPASS, SimMatch, SoftMatch, and FreeMatch showed performance comparable to our method (rank difference < CD). In contrast, other baselines, including FlexMatch, FixMatch, UDA, Π-Model, Mean Teacher, and full supervised learning, were significantly outperformed by our approach.

These results demonstrate that our method not only achieves the best average rank (1.33) but also statistically significantly outperforms 6 out of 10 competitive baselines, underscoring its effectiveness across datasets and label regimes.

H. SSL on Hyperspectral Datasets

In addition, we evaluate the proposed method on two widely used hyperspectral datasets: Indian Pines and PaviaU. For each

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:13:51 UTC from IEEE Xplore. Restrictions apply.

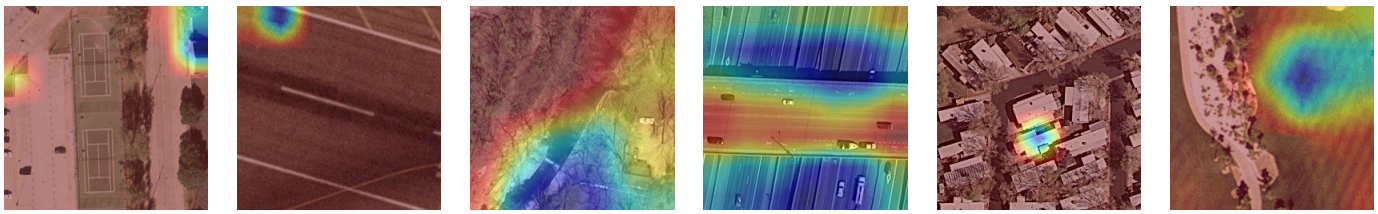


Fig. 11. Grad-CAM visualizations of representative failure cases. The model misfocuses on background or irrelevant regions in several challenging scenarios, leading to misclassification.

TABLE IV

AVERAGE RANK AND BONFERRONI–DUNN POST HOC COMPARISON RESULTS USING **Ours** AS THE BASELINE. CD = 4.39 AT $\alpha = 0.05$

Method	Avg. Rank (↓)	Rank Diff	Significant (BD)
Ours	**1.33**	—	—
EPASS	3.22	1.89	No
SimMatch	3.56	2.22	No
SoftMatch	4.11	2.78	No
FreeMatch	4.44	3.11	No
FlexMatch	6.39	5.06	**Yes**
FixMatch	6.94	5.61	**Yes**
UDA	7.33	6.00	**Yes**
Π-Model	8.44	7.11	**Yes**
Mean Teacher	9.44	8.11	**Yes**
Full Supervised	10.78	9.44	**Yes**

TABLE V

PERFORMANCE COMPARISON OF SEMI-SUPERVISED METHODS ON THE INDIAN PINES AND PAVIAU DATASETS

Indian Pines	**OA (%)**	**AA (%)**	**Kappa**
Mean Teacher	45.82 ± 1.05	56.66 ± 3.04	0.3814 ± 0.0076
FixMatch	54.11 ± 3.08	**65.47** ± 1.91	0.4798 ± 0.0324
Ours	**68.39** ± 5.38	60.55 ± 3.71	**0.5747** ± 0.0639
PaviaU	**OA (%)**	**AA (%)**	**Kappa**
Mean Teacher	65.14 ± 3.26	60.85 ± 6.60	0.5340 ± 0.0292
FixMatch	69.16 ± 2.42	67.07 ± 3.11	0.5954 ± 0.0297
Ours	**74.83** ± 3.48	**70.94** ± 2.50	**0.6618** ± 0.0508

dataset, we randomly split the entire labeled samples into training and validation subsets with a ratio of 9:1. To simulate a semi-supervised setting, we select 5 labeled samples per class from the training set and treat the remaining as unlabeled data. This setup reflects the realistic constraints of limited annotations in hyperspectral image classification.

As shown in Table V, our method consistently outperforms both FixMatch and Mean Teacher baselines across most evaluation metrics. Specifically, on the PaviaU dataset, our approach achieves the highest OA of 74.83%, average accuracy (AA) of 70.94%, and Kappa coefficient of 0.6618, surpassing FixMatch by a margin of 5.67% in OA and 3.87% in AA. Similarly, on the Indian Pines dataset, our method yields a substantial improvement in OA, reaching 68.39%, which is 14.28% higher than FixMatch and 22.57% higher than Mean Teacher.

Although our method achieves superior results in OA and Kappa, its AA on certain datasets is slightly lower than that of FixMatch. This is primarily because FixMatch tends to overfit on minority classes due to the use of strong confidence thresholds, which can sometimes lead to artificially inflated AA values, despite lower overall performance. In contrast, our method emphasizes balanced learning and overall robustness, which may sacrifice per-class accuracy uniformity to some extent in favor of better generalization.

These results demonstrate that our proposed strategy is more robust and effective under limited annotation conditions. We attribute this superior performance to the integration of confidence-aware pseudo-labeling, adaptive consistency weighting, and teacher–student decoupling, which collectively enhance the model's ability to leverage unlabeled data effectively in hyperspectral classification.

V. CONCLUSION

This study presents LCMatch, a semi-supervised remote sensing scene classification framework that builds upon FixMatch by enhancing the hierarchical structure of the pseudo-label generation process. The proposed framework integrates three key components: HCRC, label alignment, and an adaptive weighting mechanism. Specifically, HCRC randomly fuses feature representations from different layers of the primary and EMA models to improve the semantic quality and robustness of pseudo-labels. These pseudo-labels are further refined through the label alignment module to mitigate class imbalance and category confusion. Finally, the Adaptive Weighting Mechanism dynamically adjusts the loss contributions of labeled and unlabeled samples to ensure stable and effective training.

Extensive experiments on benchmark datasets demonstrate that LCMatch consistently outperforms state-of-the-art semi-supervised methods. Meanwhile, the proposed modules are lightweight and introduce minimal computational overhead, as evidenced by FLOPs, parameter counts, and runtime analysis.

Nevertheless, several limitations remain. In extremely low-label regimes (e.g., 1-shot), pseudo-label quality may degrade due to insufficient semantic guidance, which can impact model convergence. Furthermore, the method currently assumes that unlabeled data belong to the same label space as labeled data. In scenarios involving out-of-distribution samples or severe domain shifts, the current pseudo-labeling strategy may produce unreliable supervision. While our method is not specifically designed for domain adaptation, we acknowledge that robustness under domain shifts remains a critical challenge in remote sensing applications. Our approach can be seen as orthogonal to domain adaptation methods and is potentially compatible with existing domain alignment techniques such as adversarial training or feature normalization.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:13:51 UTC from IEEE Xplore. Restrictions apply.

Although preliminary experiments show limited transferability when training on UCM and testing on AID due to significant label and distribution mismatch, we consider the integration of domain adaptation modules as a promising future direction. This will be explored in subsequent studies. In addition, integrating active learning strategies may further reduce labeling cost by selecting the most informative or uncertain samples for manual annotation. This direction holds potential to enhance pseudo-label quality while maintaining label efficiency in real-world scenarios.

In future work, we plan to enhance the robustness of LCMatch by incorporating uncertainty estimation, open-set recognition, and hybrid supervision strategies. In addition to the current experiments on RGB remote sensing images, we have also conducted preliminary evaluations on hyperspectral datasets, which demonstrate the promising adaptability of LCMatch to richer spectral modalities. Moving forward, we aim to systematically extend the framework to support more complex and diverse remote sensing data, including multispectral and SAR imagery, and to explore its generalization ability under broader spectral and spatial variations. We also intend to assess the framework's deployment feasibility on resource-constrained platforms to support practical use in real-world scenarios.

Through these efforts, we hope to further improve the adaptability and generalization ability of LCMatch, contributing robust solutions for remote sensing scene classification and beyond.

REFERENCES

[1] T. Wellmann et al., "Remote sensing in urban planning: Contributions towards ecologically sound policies?," *Landscape Urban Planning*, vol. 204, Dec. 2020, Art. no. 103921.

[2] M. Weiss, F. Jacob, and G. Duveiller, "Remote sensing for agricultural applications: A meta-review," *Remote Sens. Environ.*, vol. 236, Jan. 2020, Art. no. 111402.

[3] T. R. Martha, N. Kerle, C. J. van Westen, V. Jetten, and K. V. Kumar, "Segment optimization and data-driven thresholding for knowledge-based landslide detection by object-based image analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 12, pp. 4928–4943, Dec. 2011.

[4] X. Li and G. Shao, "Object-based urban vegetation mapping with high-resolution aerial photography as a single data source," *Int. J. Remote Sens.*, vol. 34, no. 3, pp. 771–789, Feb. 2013.

[5] X. Huang, H. Liu, and L. Zhang, "Spatiotemporal detection and analysis of urban villages in mega city regions of China using high-resolution remotely sensed imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 7, pp. 3639–3657, Jul. 2015.

[6] Y. Huang, Z.-X. Chen, T. Yu, X.-Z. Huang, and X.-F. Gu, "Agricultural remote sensing big data: Management and applications," *J. Integrative Agricult.*, vol. 17, no. 9, pp. 1915–1931, Sep. 2018.

[7] S. S. Virnodkar, V. K. Pachghare, V. C. Patil, and S. K. Jha, "Remote sensing and machine learning for crop water stress determination in various crops: A critical review," *Precis. Agricult.*, vol. 21, no. 5, pp. 1121–1155, Oct. 2020.

[8] L. Dong and J. Shan, "A comprehensive review of earthquake-induced building damage detection with remote sensing techniques," *ISPRS J. Photogramm. Remote Sens.*, vol. 84, pp. 85–99, Oct. 2013.

[9] N. Rudorff, C. M. Rudorff, M. Kampel, and G. Ortiz, "Remote sensing monitoring of the impact of a major mining wastewater disaster on the turbidity of the doce river plume off the eastern Brazilian coast," *ISPRS J. Photogramm. Remote Sens.*, vol. 145, pp. 349–361, Nov. 2018.

[10] W. Wang, Y. Chen, and P. Ghamisi, "Transferring CNN with adaptive learning for remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5533918.

[11] G. Cheng, X. Xie, J. Han, L. Guo, and G.-S. Xia, "Remote sensing image scene classification meets deep learning: Challenges, methods, benchmarks, and opportunities," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 3735–3756, 2020.

[12] S. Li, W. Song, L. Fang, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Deep learning for hyperspectral image classification: An overview," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6690–6709, Sep. 2019.

[13] L. Fang, Y. Jiang, Y. Yan, J. Yue, and Y. Deng, "Hyperspectral image instance segmentation using spectral–spatial feature pyramid network," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5502613.

[14] G. J. Scott, M. R. England, W. A. Starms, R. A. Marcum, and C. H. Davis, "Training deep convolutional neural networks for land–cover classification of high-resolution imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 4, pp. 549–553, Apr. 2017.

[15] L. Wang, S. Guo, W. Huang, and Y. Qiao, "Places205-VGGNet models for scene recognition," 2015, *arXiv:1508.01667*.

[16] S.-B. Chen, Q.-S. Wei, W.-Z. Wang, J. Tang, B. Luo, and Z.-Y. Wang, "Remote sensing scene classification via multi-branch local attention network," *IEEE Trans. Image Process.*, vol. 31, pp. 99–109, 2022.

[17] Q. Zou, L. Ni, T. Zhang, and Q. Wang, "Deep learning based feature selection for remote sensing scene classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 11, pp. 2321–2325, Nov. 2015.

[18] A. M. Cheriyadat, "Unsupervised feature learning for aerial scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 1, pp. 439–451, Jan. 2014.

[19] J. Yue, L. Fang, and M. He, "Spectral–spatial latent reconstruction for open-set hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 31, pp. 5227–5241, 2022.

[20] L. Wu, L. Fang, J. Yue, B. Zhang, P. Ghamisi, and M. He, "Deep bilateral filtering network for point-supervised semantic segmentation in remote sensing images," *IEEE Trans. Image Process.*, vol. 31, pp. 7419–7434, 2022.

[21] X. Bian, C. Chen, L. Tian, and Q. Du, "Fusing local and global features for high-resolution scene classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 6, pp. 2889–2901, Jun. 2017.

[22] F. Bastani, P. Wolters, R. Gupta, J. Ferdinando, and A. Kembhavi, "SatlasPretrain: A large-scale dataset for remote sensing image understanding," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2023, pp. 16772–16782.

[23] Y. Chen, M. Mancini, X. Zhu, and Z. Akata, "Semi-supervised and unsupervised deep visual learning: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 3, pp. 1327–1347, Mar. 2024.

[24] X. Yao, L. Yang, G. Cheng, J. Han, and L. Guo, "Scene classification of high resolution remote sensing images via self-paced deep learning," in *Proc. Dig. Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jan. 2019, pp. 521–524.

[25] X. Yang, Z. Song, I. King, and Z. Xu, "A survey on deep semi-supervised learning," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 9, pp. 8934–8954, Sep. 2023.

[26] W. Han, R. Feng, L. Wang, and Y. Cheng, "A semi-supervised generative framework with deep learning features for high-resolution remote sensing image scene classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 145, pp. 23–43, Nov. 2018.

[27] W. Miao, J. Geng, and W. Jiang, "Semi-supervised remote-sensing image scene classification using representation consistency Siamese network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5616614.

[28] Y. Xiong, K. Xu, Y. Dou, Y. Zhao, and Z. Gao, "WRMatch: Improving fixmatch with weighted nuclear-norm regularization for few-shot remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5612214.

[29] L. Wu, L. Fang, X. He, M. He, J. Ma, and Z. Zhong, "Querying labeled for unlabeled: Cross-image semantic consistency guided semi-supervised semantic segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 7, pp. 8827–8844, Jul. 2023.

[30] D.-H. Lee et al., "Pseudo-label: The simple and efficient semi-supervised learning method for deep neural networks," in *Proc. Workshop Challenges Represent. Learn. (ICML)*, vol. 3, no. 2, Atlanta, GA, USA, 2013, p. 896.

[31] K. Sohn et al., "FixMatch: Simplifying semi-supervised learning with consistency and confidence," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 596–608.

[32] F. Klinker, "Exponential moving average versus moving exponential average," *Mathematische Semesterberichte*, vol. 58, no. 1, pp. 97–107, Apr. 2011.

[33] A. A. Elngar et al., "Image classification based on CNN: A survey," *J. Cybersecur. Inf. Manage.*, vol. 6, no. 1, pp. 18–50, 2021.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:13:51 UTC from IEEE Xplore. Restrictions apply.

[34] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," 2016, *arXiv:1609.02907*.
[35] M. Chen, Z. Wei, Z. Huang, B. Ding, and Y. Li, "Simple and deep graph convolutional networks," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 1725–1735.
[36] S. Zhang, H. Tong, J. Xu, and R. Maciejewski, "Graph convolutional networks: A comprehensive review," *Comput. Social Netw.*, vol. 6, no. 1, pp. 1–23, Dec. 2019.
[37] J. Fang, Y. Yuan, X. Lu, and Y. Feng, "Robust space–frequency joint representation for remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 7492–7502, Oct. 2019.
[38] B. Li et al., "Aggregated deep Fisher feature for VHR remote sensing scene classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 9, pp. 3508–3523, Sep. 2019.
[39] J. Zhang, H. Zhao, and J. Li, "TRS: Transformers for remote sensing scene classification," *Remote Sens.*, vol. 13, no. 20, p. 4143, Oct. 2021.
[40] U. Muhammad, W. Wang, S. P. Chattha, and S. Ali, "Pre-trained VGGNet architecture for remote-sensing image scene classification," in *Proc. 24th Int. Conf. Pattern Recognit. (ICPR)*, Aug. 2018, pp. 1622–1627.
[41] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[42] D. Lin, K. Fu, Y. Wang, G. Xu, and X. Sun, "MARTA GANs: Unsupervised representation learning for remote sensing image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 11, pp. 2092–2096, Nov. 2017.
[43] H. Tang, H. Liu, D. Xu, P. H. S. Torr, and N. Sebe, "AttentionGAN: Unpaired image-to-image translation using attention-guided generative adversarial networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 4, pp. 1972–1987, Apr. 2023.
[44] D. Ma, P. Tang, and L. Zhao, "SiftingGAN: Generating and sifting labeled samples to improve the remote sensing image scene classification baseline in vitro," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 7, pp. 1046–1050, Jul. 2019.
[45] Y. Li et al., "MEET: A million-scale dataset for fine-grained geospatial scene classification with zoom-free remote sensing imagery," *IEEE/CAA J. Automatica Sinica*, vol. 12, no. 5, pp. 1004–1023, May 2025.
[46] C.-I. Chang, C.-C. Liang, P.-C. Chung, and P. F. Hu, "Hierarchical one-class detection for hyperspectral image classification with background," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2024, Art. no. 5500227.
[47] S. Laine and T. Aila, "Temporal ensembling for semi-supervised learning," 2016, *arXiv:1610.02242*.
[48] A. Tarvainen and H. Valpola, "Mean teachers are better role models: Weight-averaged consistency targets improve semi-supervised deep learning results," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, p. 1196.
[49] T. Miyato, S.-I. Maeda, M. Koyama, and S. Ishii, "Virtual adversarial training: A regularization method for supervised and semi-supervised learning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 41, no. 8, pp. 1979–1993, Aug. 2019.
[50] Q. Xie, Z. Dai, E. Hovy, M.-T. Luong, and Q. V. Le, "Unsupervised data augmentation for consistency training," in *Proc. NIPS*, vol. 33, 2020, pp. 6256–6268.
[51] Q. Xie, M.-T. Luong, E. Hovy, and Q. V. Le, "Self-training with noisy student improves ImageNet classification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2020, pp. 10687–10698.
[52] B. Zhang et al., "FlexMatch: Boosting semi-supervised learning with curriculum pseudo labeling," in *Proc. Adv. Neural Inf. Process. Syst.*, 2021, pp. 18408–18419.
[53] Y. Wang, G. Liu, L. Yang, J. Liu, and L. Wei, "An attention-based feature processing method for cross-domain hyperspectral image classification," *IEEE Signal Process. Lett.*, vol. 32, pp. 196–200, 2025.
[54] S. Liu, C. Fu, Y. Duan, X. Wang, and F. Luo, "Spatial-spectral enhancement and fusion network for hyperspectral image classification with few labeled samples," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2024, Art. no. 5502414.
[55] G.-S. Xia et al., "AID: A benchmark data set for performance evaluation of aerial scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3965–3981, Jul. 2017.
[56] Y. Yang and S. Newsam, "Bag-of-visual-words and spatial extensions for land-use classification," in *Proc. 18th SIGSPATIAL Int. Conf. Adv. Geograph. Inf. Syst.*, Nov. 2010, pp. 270–279.
[57] G. Cheng, J. Han, and X. Lu, "Remote sensing image scene classification: Benchmark and state of the art," *Proc. IEEE*, vol. 105, no. 10, pp. 1865–1883, Oct. 2017.
[58] Y. Wang et al., "USB: A unified semi-supervised learning benchmark for classification," in *Proc. 36th Conf. Neural Inf. Process. Syst. Datasets Benchmarks Track*, 2022, pp. 3938–3961.
[59] M. Zheng, S. You, L. Huang, F. Wang, C. Qian, and C. Xu, "SimMatch: Semi-supervised learning with similarity matching," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 14471–14481.
[60] Y. Wang et al., "FreeMatch: Self-adaptive thresholding for semi-supervised learning," 2022, *arXiv:2205.07246*.
[61] H. Chen et al., "SoftMatch: Addressing the quantity-quality trade-off in semi-supervised learning," 2023, *arXiv:2301.10921*.
[62] K.-B. Nguyen, "Debiasing, calibrating, and improving semi-supervised learning performance via simple ensemble projector," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2024, pp. 2441–2451.
[63] M. C. Cieslak, A. M. Castelfranco, V. Roncalli, P. H. Lenz, and D. K. Hartline, "T-distributed stochastic neighbor embedding (t-SNE): A tool for eco-physiological transcriptomic analysis," *Mar. Genomics*, vol. 51, Jun. 2020, Art. no. 100723.
[64] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 618–626.



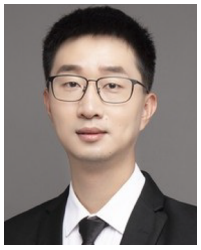
Ruizhe Hu received the B.E. degree in software engineering from Fujian Agriculture and Forestry University, Fuzhou, China, in 2022, and the M.S. degree in computer science from Fujian University of Technology, Fuzhou, in 2025.

He is currently a Faculty Member of Fujian Provincial Key Laboratory of Information Processing and Intelligent Control, School of Computer and Data Science, Minjiang University, Fuzhou. His research interests include remote sensing scene understanding, computer vision, and semi-supervised learning (SSL).



Zuoyong Li (Senior Member, IEEE) received the B.E. and M.S. degrees in computer science and technology from Fuzhou University, Fuzhou, China, in 2002 and 2006, respectively, and the Ph.D. degree from the School of Computer Science and Technology, Nanjing University of Science and Technology, Nanjing, China, in 2010.

He is currently a Professor with the School of Computer and Data Science, Minjiang University, Fuzhou. He has published over 100 articles in international/national journals. His research interests include image processing, pattern recognition, and machine learning.



Tao Wang (Member, IEEE) received the B.E. degree in information engineering from South China University of Technology, Guangzhou, China, in 2009, and the Ph.D. degree in computer science from The Australian National University, Canberra, ACT, Australia, in 2016.

He was a member of the Computer Vision Research Group, National ICT Australia, Canberra. He is currently an Associate Professor with the School of Computer and Data Science, Minjiang University, Fuzhou, China. His research interests include scene understanding, object detection, and semantic instance segmentation.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:13:51 UTC from IEEE Xplore. Restrictions apply.



Sien Li received the B.E. degree in software engineering from Minjiang University, Fuzhou, China, in 2022, and the M.S. degree in computer science from Fuzhou University, Fuzhou, in 2025.

His research interests include computer vision, pattern recognition, and semi-supervised learning (SSL).



Yuanzheng Cai received the B.E. degree in software engineering from Fujian Normal University, Fuzhou, China, in 2010, the M.S. degree in computer science from Yunnan University, Kunming, China, in 2012, and the Ph.D. degree in artificial intelligence from Xiamen University, Xiamen, China, in 2016.

He is currently a Lecturer with the School of Computer and Data Science, Minjiang University, Fuzhou. His research interests include image/video retrieval, machine learning, and object recognition.



Rong Hu received the M.S. degree in computer science and technology from Changsha University of Science and Technology, Changsha, China, in 2006, and the Ph.D. degree from the School of Computer Science and Technology, Nanjing University of Science and Technology, Nanjing, China, in 2013.

She is currently a Professor with the School of Computer Science and Mathematics, Fujian University of Technology, Fuzhou, China. She has published more than 30 articles in international and national journals. Her research interests include image processing, artificial intelligence, and machine learning.



George N. Papageorgiou (Member, IEEE) received the Bachelor of Engineering (B.Eng.) degree (Hons.) from the City University of London, London, U.K., in 1992, the master's degree in business administration (MBA) from the University of Kent, Canterbury, U.K., in 1994, and the Doctor of Philosophy (Ph.D.) degree from the City University of London, in 1997.

He is currently a Full Professor with European University Cyprus (EUC), Nicosia, Cyprus, where he is also the Director of the SYSTEMA Research Centre, and he is a voting member of the European Committee for Standardization CEN/ISO. He has over 30 years of experience as an educator, consultant, and researcher participating in numerous European-funded projects as a coordinator and research associate. Furthermore, he has held positions as a visiting professor at universities in Europe, China, and the USA. He has published over 100 peer-reviewed papers in renowned scientific journals and conference proceedings.

Prof. Papageorgiou has served as a member of numerous academic and professional societies, including the IEEE, and has been elected as president of the Cyprus IEEE Intelligent Transportation Systems Society (ITSS).

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:13:51 UTC from IEEE Xplore. Restrictions apply.